

PYTHON

Module 21: Sequence Iteration & For Loops

CODING

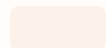
Sequence Iteration with For Loops

A `for` loop in Python is used for iterating over a sequence—be it a list, tuple, dictionary, set, or string.

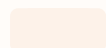
Functional Shift at CodeHive:



- ****Iterator Logic****: Unlike C-style `for` loops, Python's `for` works like an iterator method, visiting each item in a collection directly.
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- ****No Counter Required****: You don't need to manage an index variable (like `i`) unless you specifically choose to use the `range()` function.
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The range() Function

When you need to iterate a specific number of times rather than over a collection, `range()` is the standard tool.

- **Parameters**: `range(start, stop, step)`
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- **Range Behavior**: `range(6)` generates 0 to 5. The 'stop' value is always exclusive.
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- **Step Control**: Use the third parameter to increment by values other than 1 (e.g., `range(0, 10, 2)` for even numbers).
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Control Statements: break & continue

- ****break****: Stops the loop entirely before it has finished visiting all items.
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- ****continue****: Immediately ends the current item's processing and moves to the next item in the sequence.
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Example: Use

break to exit a search once the target is found to save system resources.

The Loop Completion 'else'

The `else` keyword in a `for` loop executes a block of code strictly when the loop finishes its entire sequence.

****The Critical Rule**:** If the loop is terminated by a `break`, the `else` block is ignored. This makes it perfect for 'not found' logic in search algorithms.

Nested Loops & Complexity

Nesting occurs when a for loop is placed inside another. The inner loop completes all its iterations for every single iteration of the outer loop.

****CodeHive Warning****: Be cautious with deep nesting as it can exponentially increase the time complexity ($O(n^2)$) of your script.

The Null Iterator (pass)

Like
if statements,
for loops cannot be syntactically empty. If you need a
loop that does nothing (as a placeholder), use the
pass statement to avoid syntax errors.